



LLM CONDUCTOR ORCHESTRATION MODEL

MUL

Mul AI

A small conductor model that assembles and directs the best AI team for each task, behind a single API

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Mul means water in Korean, a counterpart to Sakana (fish). Inspired by Japan's Sakana Fugu architecture.



DEFINITION

What is Mul AI

Not a bigger model. A smarter conductor.

A small conductor model takes your request and either answers directly or assembles a team of specialist models, splits the work, verifies it, then merges everything into one answer. All of this is delivered through a single OpenAI compatible API, so you call it just like one model.

True to its name, Mul flows to fit the shape of each problem and forms the answer that fits best.

Why it is needed

01

The limits of one model

No single large model can be the best at every kind of task.

02

Uneven strengths

Coding, reasoning, Korean, long context. Each model is good at different things.

03

Tedious switching

Swapping models for every task is cumbersome and inefficient.

04

Exploding complexity

Wiring many models together yourself quickly raises cost and operational burden.

Intelligence lives in the orchestration

The core of the product is not bigger weights, but how the pool is assembled and coordinated. Intelligence comes from how the models are connected, not from inside any single one.

A small conductor

One light, fast model directs the whole flow.

Swappable players

Specialist models are pool members you add or remove by config.

Strategy is the edge

What gets routed where is the differentiator.

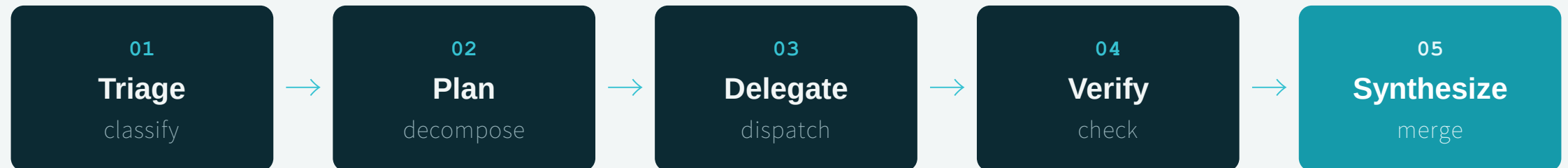
Two paths, one answer

Fast path

Simple requests are answered directly by the conductor for a quick response.

Orchestration path

Complex requests assemble a team of specialists to divide the work.



Three roles, plus recursion

THINKER

Reason

Breaks the problem down and sets the reasoning strategy and order.

WORKER

Execute

The best suited model actually performs each assigned subtask.

VERIFIER

Check

Checks the accuracy and quality of intermediate results and filters out errors.

Recursion

For a hard sub-problem, the conductor can spin up another conductor and form a smaller team inside it. Depth is safely capped.

A Korean first worker pool

Korean models

Upstage Solar Pro

LG EXAONE

Naver HyperCLOVA X

Global frontier

OpenAI

Anthropic

Google

Open weight

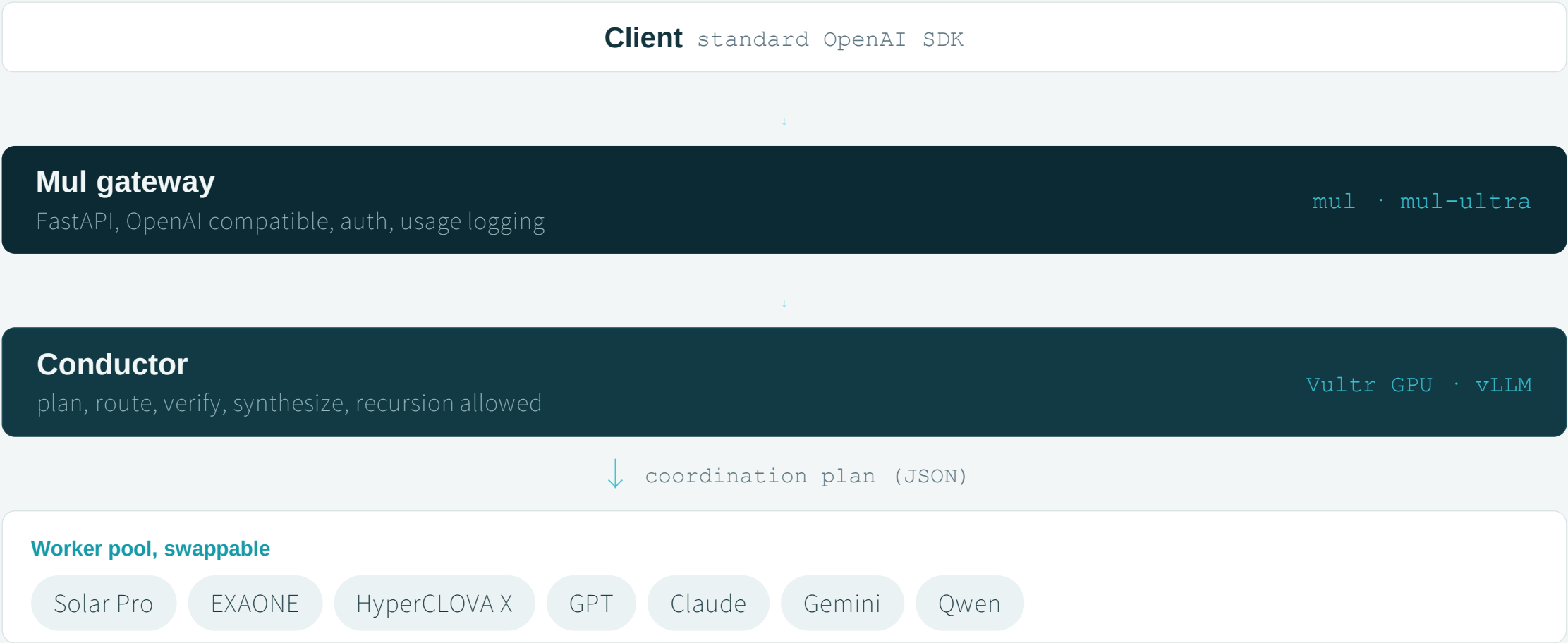
Qwen

Llama

Self hostable

Every model is swappable through a config file, and specific providers can be excluded per request or per account for data and compliance needs.

The path a request flows



What it is built with

Layer	Components
Gateway	FastAPI, OpenAI compatible API, streaming
Conductor serving	vLLM, open weight model (GPU)
Worker adapters	Async client per provider (httpx)
Orchestration	asyncio, parallel dispatch, safety caps
Frontend	Next.js, React, TypeScript, Tailwind
Data	SQLite (dev) to Postgres (prod)

No heavy agent framework. The orchestration logic is the product, so it is written explicitly to stay legible and tunable.

Vultr GPU deployment

Core principle

Only the conductor runs on the GPU. Workers call external APIs, so the rented GPU box stays small and cheap.

Recommended spec

NVIDIA L40S 48GB or A100 80GB

Serving

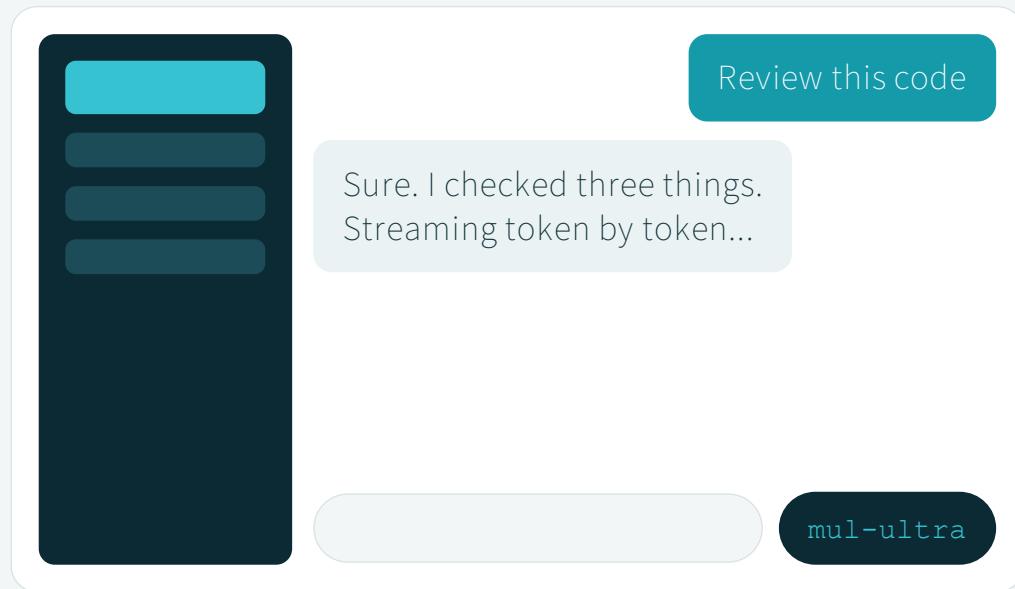
Local serving via a vLLM OpenAI compatible server

Flexibility

GPU and conductor model swap via environment variables

Note: L40S runs about US\$0.85 to 1.67 per hour and A100 80GB about US\$1.29. These are approximate and vary by region and time.

A familiar experience with its own face



- ◆ A familiar experience like Claude or ChatGPT, but with Mul's own design
- ◆ Korean first UI with token by token streaming output
- ◆ Conversation sidebar, code block copy, mul / mul-ultra switch
- ◆ Admin only orchestration viewer: see which roles collaborated and how

Roadmap and current status

R&D

1 Gateway + fast path

2 Worker registry + adapters

3 Orchestration loop

4 Billing + traces

5 Chat UI (Next.js)

6 Eval harness

7 Learned conductor (research)

Phases 1 to 5 are the shippable product. Phase 7 is research into evolutionary and RL based routing.



Like water, it flows to fit the shape of the problem.

A Korean orchestration model where a small conductor directs the best AI team.

Mul AI · Water

